

IOWA STATE UNIVERSITY

LAB 4 - SEMI-TENSION FIELD BEAM REPORT

LUCAS TAVARES VASCONCELLOS

PROFESSOR

DR. VINAY DAYAL

College of Engineering

Aerospace Engineering

Aer E 4260 Design of Aerospace Structures

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INTRODUCTION

In Lab 4, we are tasked with designing the built-up main spar of an aircraft wing. The design parameters, including specific dimensions and material selection of aluminum 2024-T3, form the basis of our analysis. This problem offers an open-ended approach to designing the spar, allowing for flexibility in determining the skin, stiffeners, and flanges.

The goal is to create a structurally sound spar, ensuring that key elements such as the stiffeners and chords do not buckle under load while allowing for controlled buckling in the skin. Additionally, the stresses in all components must remain within the material yield limits. A factor of safety of 1.5 must be maintained throughout the design to account for uncertainties and ensure reliability. The final design must also incorporate rivet patterns that ensure proper load transfer, with calculations provided for both horizontal and vertical members.

FIGURES AND TABLES

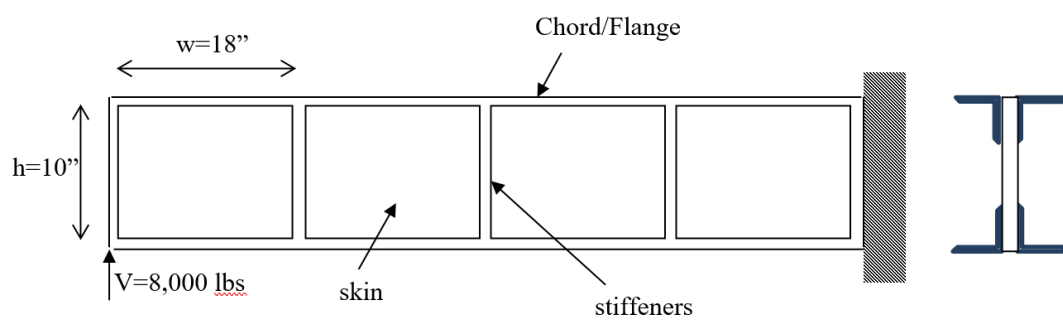


Figure 2.1: Side and front view of beam

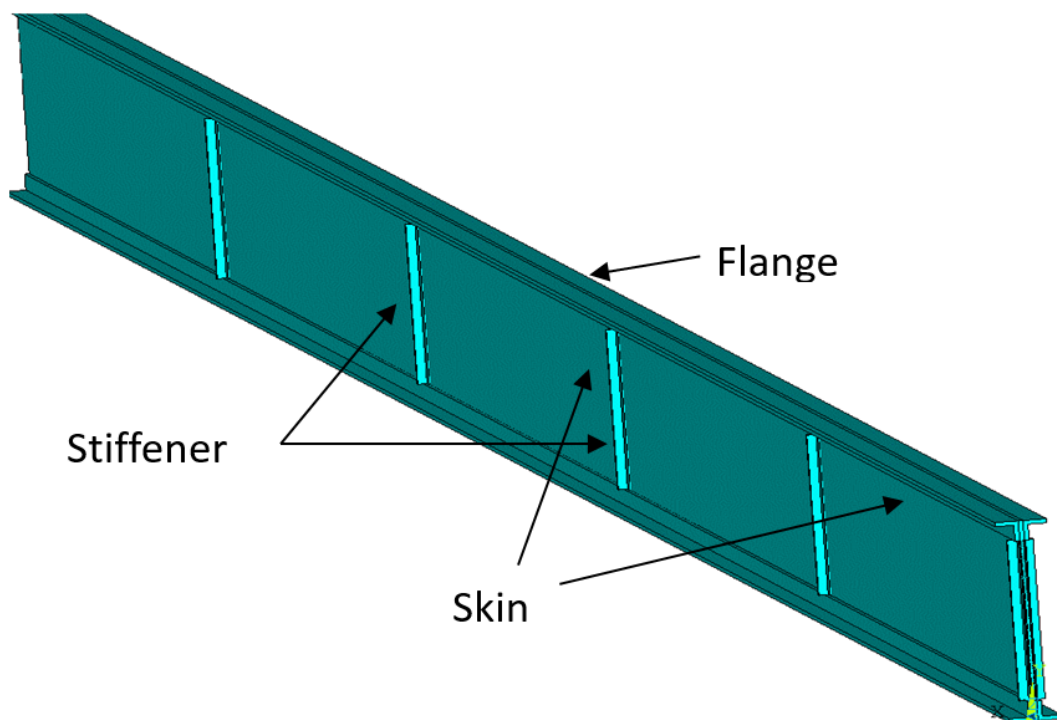


Figure 2.2: Isometric view of beam

3-74

Table 3.2.3.0(e ₁). Design Mechanical and Physical Properties of Clad 2024 Aluminum Alloy Sheet and Plate																				
Specification	AMS-QQ-A-250/5																			
Form	Flat sheet and plate																			
Temper	T3								T351											
Thickness, in.	0.008-0.009		0.010-0.062		0.063-0.128		0.129-0.249		0.250-0.499		0.500-1.000 ^a		1.001-1.500 ^a		1.501-2.000 ^a		2.001-3.000 ^a		3.001-4.000 ^a	
Basis	A	B	A	B	A	B	A	B	A	B	A	B	A	B	A	B	A	B	A	B
Mechanical Properties:																				
F_{br} , ksi:																				
L	59	60	60	61	62	63	63	64	62	64	61	63	60	62	60	62	58	60	55	57
LT	58	59	59	60	61	62	62	63	62	64	61	63	60	62	60	62	58	60	55	57
ST	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	52 ^b	54 ^b	49 ^b	51 ^b
F_{br} , ksi:																				
L	44	45	44	45	45	47	45	47	46	48	45	48	45	48	45	47	44	46	39	41
LT	39	40	39	40	40	42	40	42	40	42	40	42	40	42	40	42	40	42	39	41
ST	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	38 ^b	40 ^b	38 ^b	39 ^b
F_{br} , ksi:																				
L	36	37	36	37	37	39	37	39	37	39	37	39	37	39	36	38	35	37	33	35
LT	42	43	42	43	43	45	43	45	43	45	42	45	42	44	42	44	41	43	39	41
ST	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	46	48	44	47
F_{br} , ksi:																				
L	37	37	37	38	38	39	39	40	37	38	36	37	35	37	35	37	34	35	32	34
F_{br} , ksi:																				
(e/D = 1.5)	96	97	97	99	101	102	102	104	94	97	92	95	91	94	91	94	88	91	83	86
(e/D = 2.0)	119	121	121	123	125	127	127	129	115	119	113	117	111	115	111	115	107	111	102	106
F_{br} , ksi:																				
(e/D = 1.5)	68	70	68	70	70	73	70	73	69	72	69	72	69	72	69	72	69	72	67	70
(e/D = 2.0)	82	84	82	84	84	88	84	88	82	86	82	86	82	86	82	86	82	86	80	84
ϵ , percent (S-basis):																				
LT	10	...	d	...	15	...	15	...	12	...	8	...	7	...	6	...	4	...	4	...
E , 10 ³ ksi:																				
Primary	10.5								10.7											
Secondary	9.5				10.0				10.2											
E , 10 ³ ksi:																				
Primary	10.7								10.9											
Secondary	9.7				10.2				10.4											
G , 10 ³ ksi	...																			
μ	0.33																			
Physical Properties:																				
α , lb/in. ³	0.100																			
C , K , and α	...																			

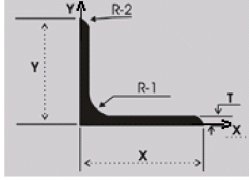
a These values, except in the ST direction, have been adjusted to represent the average properties across the whole section, including the 2-½ percent nominal cladding thickness.

b Caution: This specific alloy, temper, and product form exhibits poor stress-corrosion cracking resistance in this grain direction. It corresponds to an SCC resistance rating of D, as indicated in Table 3.1.2.3.1(a).

c Bearing values are “dry pin” values per Section 1.4.7.1. See Table 3.1.2.1.1.

d See Table 3.2.3.0(f).

Table 2.1: Design Mechanical and Physical Properties of Clad 2024 Aluminum Alloy Sheet and Plate



For an angle beam the properties are

0	NO.	X	Y	T	R-1	R-2	WT/FT	A	lxx	lyy	ybar	xbar	lxx/cx	lyy/cy
1	16394	1	1.5	0.156	0.078	0.015	0.439	0.39	0.1768	0.0539	0.4763	0.2419	0.0863	0.041
2	16131	1	1	0.188	0.125	0.094	0.408	0.376	0.0649	0.0649	0.2882	0.2882	0.0473	0.0473
3	19120	1	1	0.25	0.125	0.093	0.533	0.5	0.0885	0.0885	0.2969	0.2969	0.0632	0.0632
4	18405	1.125	1.125	0.094	0.188	0.094	0.248	0.2115	0.0449	0.0449	0.3028	0.3028	0.031	0.031
5	18832	1.25	1.625	0.188	0.188	0.188	0.598	0.5405	0.2717	0.126	0.494	0.3187	0.1236	0.0763
6	16442	1.25	1.25	0.25	0.25	0.125	0.683	0.625	0.1693	0.1693	0.3625	0.3625	0.0982	0.0982
7	230	1.25	1.75	0.188	0.188	0.125	0.634	0.564	0.3386	0.1263	0.5437	0.3094	0.1425	0.0769
8	18309	1.5	1.5	0.125	0.188	0.125	0.432	0.375	0.1416	0.1416	0.4036	0.4036	0.0734	0.0734
9	18902	1.5	1.5	0.25	0.188	0.125	0.826	0.75	0.2891	0.2891	0.4271	0.4271	0.1419	0.1419
10	18707	1.5	2.5	0.313	0.25	0.03	1.366	1.252	1.6455	0.3777	0.8277	0.3668	0.4711	0.1846
11	18654	1.75	1.25	0.125	0.188	0.125	0.432	0.375	0.0825	0.2241	0.2943	0.5339	0.0523	0.0964
12	17830	1.75	1.75	0.25	0.187	0.125	0.956	0.875	0.4557	0.4557	0.4911	0.4911	0.1944	0.1944
13	19304	2	2	0.309	0.25	0.125	1.376	1.236	0.8437	0.8437	0.5653	0.5653	0.3128	0.3128
14	19305	2	2	0.185	0.25	0.125	0.854	0.74	0.4976	0.4976	0.542	0.542	0.1922	0.1922
15	18702	2	2	0.125	0.125	0.015	0.586	0.5	0.3346	0.3346	0.5293	0.5293	0.1323	0.1323
16	18904	2	2	0.125	0.25	0.125	0.589	0.5	0.3346	0.3346	0.5293	0.5293	0.1323	0.1323
17	18391	2	2.5	0.188	0.25	0.125	0.98	0.846	0.9836	0.5069	0.7323	0.4927	0.2998	0.2
18	16134	2	2	0.25	0.25	0.125	1.133	1	0.6771	0.6771	0.5547	0.5547	0.2556	0.2556
19	18708	2	3	0.38	0.38	0.038	2.081	1.9	3.4566	1.0682	0.9616	0.4996	0.8339	0.3959
20	16302	2.25	3.5	0.25	0.312	0.03	1.674	1.4375	3.5846	0.9674	1.1087	0.5109	0.7601	0.3405
21	16133	2.5	2.5	0.25	0.25	0.125	1.433	1.25	1.3151	1.3151	0.6813	0.6813	0.4041	0.4041
22	17672	3	3	0.188	0.312	0.1	1.301	1.128	1.6986	1.6986	0.7941	0.7941	0.4476	0.4476
23	16443	3	2	0.25	0.25	0.125	1.4	1.25	0.6823	2.2604	0.4688	0.9438	0.2662	0.5578
24	18246	3	3	0.25	0.312	0.25	1.718	1.5	2.2656	2.2656	0.8073	0.8073	0.5874	0.5874
25	18703	3	2	0.313	0.313	0.313	1.733	1.565	0.8653	2.8374	0.4841	0.9528	0.3289	0.692
26	16132	3	3	0.375	0.312	0.25	2.524	2.25	3.4277	3.4277	0.832	0.832	0.8626	0.8626
27	17650	3	3.5	0.5	0.375	0.25	3.532	3.25	7.2708	4.6458	1.0385	0.8077	1.5299	1.152
28	16935	3.5	1.5	0.25	0.25	0.06	1.5	1.25	0.2995	3.5807	0.3063	1.2563	0.1527	0.7167
29	16143	3.5	3.5	0.375	0.375	0.25	2.986	2.625	5.4209	5.4209	0.9587	0.9587	1.1837	1.1837
30	17788	3.5	5	0.375	0.437	0.312	3.582	3.1875	15.6865	5.4473	1.5395	0.8226	2.3499	1.2289
31	17789	3.5	3.5	0.5	0.375	0.25	3.826	3.5	7.2917	7.2917	0.9821	0.9821	1.5551	1.5551
32	17651	3.5	5	0.5	0.437	0.312	4.703	4.25	20.9792	7.3542	1.5588	0.8529	3.0955	1.6102
33	18490	8	6	0.625	0.5	0.312	10.05	8.75	45.651	107.1549	1.4503	2.4057	5.9885	10.1023

Table 2.2: Angle Beam Mechanical and Physical Properties

Fastener Diameter		3/32	1/8	5/32	3/16	1/4	5/16	3/8	7/16	1/2	9/16
Fastener Material	F_u (Ksi)	Ultimate Single Shear Load (lb/fastener)									
1100F	9										
5056(B)	28	203	363	556	802	1450	2290	3280			
2117-T3(AD)	30	217	388	596	862	1550	2460	3510			
2017-T31(D)	34	247	442	675	977	1760	2790	3970			
2017-T3(D)	38	275	494	755	1090	1970	3110	4450			
2024-T31(DD)	41	296	531	815	1180	2120	3360	4800			
Monel	49	355	635	973	1400	2540	4020	5730			
7075-H75	38	275	494	755	1090	1970	3110	4450			
A-286 CRES	90	—	—	1726	2485	4415	6906	9938			
Ti-6Al-4V & Alloy Steel	95	—	—	1822	2623	4660	7290	10490	14280	18650	23610
Alloy Steel	108	—	—	2071	2982	5300	8280	11930	16240	21210	26840
Alloy Steel	125	—	—	2397	3452	6140	9590	13810	18790	24540	31060
H-11 Steel	132	—	—	2531	3645	6480	10120	14580	19840	25920	32800

(Refer to MIL-HDBK-5D per Ref. 7.6)

Fig. 7.1.3 Fastener shear-off allowable loads.

Table 2.3: Fastener shear-off allowable loads

METHODOLOGY

3.1 Skin Design

Starting with the skin, the first step was to calculate the skin thickness. The thickness can be obtained from the shear stress formula using the maximum shear load (with a factor of safety of 1.5) and the maximum shear stress (from [Table 2.1](#)):

$$t_{\min} = \frac{V_{\max}}{h \cdot \tau_{\max}} = \frac{12 \text{ kip}}{(10 \text{ in}) \cdot (37 \text{ ksi})} = 0.0324 \text{ in} \quad (3.1)$$

Since aluminum sheets are manufactured in multiples of 1/64 in, our thickness shall be:

$$t = \frac{3}{64} \text{ in} = 0.046875 \text{ in} \quad (3.2)$$

With the web thickness defined, we can calculate the actual shear stress with the maximum shear load:

$$\tau = \frac{8 \text{ kip}}{(10 \text{ in}) \cdot \frac{3}{64} \text{ in}} = 17.067 \text{ ksi} \quad (3.3)$$

and the maximum shear stress:

$$\tau_{\max} = \frac{12 \text{ kip}}{(10 \text{ in}) \cdot \frac{3}{64} \text{ in}} = 25.6 \text{ ksi} \quad (3.4)$$

Thus, the factor of safety with respect to shear stress for the skin is:

$$FS_{\tau} = \frac{37 \text{ ksi}}{17.067 \text{ ksi}} = 2.17 \quad (3.5)$$

3.2 Stiffener Design

For the stiffeners, we start by calculating the minimum area and moment of inertia required to select a suitable beam from [Table 2.2](#). Using the yield compressive stress of 36 ksi and Young's modulus of 10,700 ksi (from [Table 2.1](#)), we calculate the maximum and actual compressive loads:

$$P_{\max} = (12 \text{ kip}) \cdot \left(\frac{18 \text{ in}}{10 \text{ in}} \right) \cdot \tan(42.5^\circ) = 19.793 \text{ kip} \quad (3.6)$$

$$P = (8 \text{ kip}) \cdot \left(\frac{18 \text{ in}}{10 \text{ in}} \right) \cdot \tan(42.5^\circ) = 13.195 \text{ kip} \quad (3.7)$$

With $P_{\max}$, we calculate the minimum area $A_{\min}$ and moment of inertia $I_{\min}$:

$$A_{\min} = \frac{P_{\max}}{\sigma_{c,y}} = \frac{19.793 \text{ kip}}{36 \text{ ksi}} = 0.550 \text{ in}^2 \quad (3.8)$$

$$I_{\min} = \frac{P_{\max} \cdot (10 \text{ in})^2}{\pi^2 \cdot (10.7 \times 10^3 \text{ ksi})} = 0.0187 \text{ in}^4 \quad (3.9)$$

From Table 2.2, beam number 2 was chosen with $A = 0.376 \text{ in}^2$ and $I = 0.0649 \text{ in}^4$. Since there are two stiffeners, the total area and moment of inertia are:

$$A = 2 \times 0.376 \text{ in}^2 = 0.752 \text{ in}^2 \quad (3.10)$$

$$I = 2 \times 0.0649 \text{ in}^4 = 0.1298 \text{ in}^4 \quad (3.11)$$

The compressive stress is then:

$$\sigma_c = \frac{13.195 \text{ kip}}{0.752 \text{ in}^2} = 17.547 \text{ ksi} \quad (3.12)$$

with a factor of safety:

$$FS_{\sigma_c} = \frac{36 \text{ ksi}}{17.547 \text{ ksi}} = 2.05 \quad (3.13)$$

The critical buckling load is:

$$P_{\text{cr}} = \frac{\pi^2 \cdot (10.7 \times 10^3 \text{ ksi}) \cdot (0.1298 \text{ in}^4)}{(10 \text{ in})^2} = 137.075 \text{ kip} \quad (3.14)$$

Thus, the factor of safety for buckling is:

$$FS_{\text{buckling}} = \frac{P_{\text{cr}}}{P} = \frac{137.075 \text{ kip}}{13.195 \text{ kip}} = 10.39 \quad (3.15)$$

3.3 Flange Design

For the flange, the beam selection is based on I/c and A derived from the maximum compressive and tensile stresses. First, the maximum moment is calculated:

$$M_{\max} = (25.6 \text{ ksi}) \cdot \tan(42.5^\circ) \cdot \left(\frac{3}{64} \text{ in} \right) \cdot \left(\frac{18 \text{ in}}{12} \right)^2 = 29.689 \text{ kip} \cdot \text{in} \quad (3.16)$$

Using the yield compressive and tensile stresses, we develop the following system of equations to solve for $(I/c)_{\min}$ and $A_{\min}$:

$$\sigma_{C,y} = -36 \text{ ksi} = -\frac{M_{\max}}{I/c} + \frac{12 \text{ kip} \cdot \cot(42.5^\circ)}{2A} \quad (3.17)$$

$$\sigma_{T,y} = 44 \text{ ksi} = \frac{M_{\max}}{I/c} + \frac{12 \text{ kip} \cdot \cot(42.5^\circ)}{2A} \quad (3.18)$$

Solving these, we obtain:

$$\left(\frac{I}{c}\right)_{\min} = 0.742 \text{ in}^3, A_{\min} = 1.636 \text{ in}^2 \quad (3.19)$$

Since there are two flanges, we can divide those values by two:

$$\left(\frac{I}{c}\right)_{\min} = 0.371 \text{ in}^3, A_{\min} = 0.818 \text{ in}^2 \quad (3.20)$$

Beam number 21 from Table 2.2 was chosen, with $A = 1.25 \text{ in}^2$, $I/c = 0.4041 \text{ in}^3$, and $I = 1.3151 \text{ in}^4$. The actual moment was calculated, and with the chosen beam properties, the actual compressive and tensile stresses are:

$$\sigma_T = \frac{M}{2 \cdot I/c} + \frac{P}{2 \cdot 2A} = \frac{19.793 \text{ kip} \cdot \text{in}}{2 \cdot 0.4041 \text{ in}^3} + \frac{8 \text{ kip} \cdot \cot(42.5^\circ)}{2 \cdot 2 \cdot 1.25 \text{ in}^2} = 26.236 \text{ ksi} \quad (3.21)$$

$$\sigma_C = -\frac{M}{2 \cdot I/c} + \frac{P}{2 \cdot 2A} = -\frac{19.793 \text{ kip} \cdot \text{in}}{2 \cdot 0.4041 \text{ in}^3} + \frac{8 \text{ kip} \cdot \cot(42.5^\circ)}{2 \cdot 2 \cdot 1.25 \text{ in}^2} = -22.744 \text{ ksi} \quad (3.22)$$

The load and critical load for buckling are calculated as follows:

$$P = \sigma_C \cdot 2A = (45.488 \text{ ksi}) \cdot 2 \cdot (1.25 \text{ in}^2) = 113.721 \text{ kip} \quad (3.23)$$

$$P_{\text{cr}} = \frac{\pi^2 \cdot (10.7 \times 10^3 \text{ ksi}) \cdot 2 \cdot (1.3151 \text{ in}^4)}{(18 \text{ in})^2} = 857.289 \text{ kip} \quad (3.24)$$

Finally, the factors of safety are:

$$FS_{\sigma_T} = \frac{44 \text{ ksi}}{26.236 \text{ ksi}} = 1.68 \quad (3.25)$$

$$FS_{\sigma_C} = \frac{36 \text{ ksi}}{22.744 \text{ ksi}} = 1.58 \quad (3.26)$$

$$FS_{\text{buckling}} = \frac{857.289 \text{ kip}}{113.721 \text{ kip}} = 7.54 \quad (3.27)$$

3.4 Rivets Design

We begin by calculating the shear flow q and $q_{\max}$, based on the shear stress from the maximum shear load (with a factor of safety of 1.5):

$$q = \tau \cdot t = (17.067 \text{ ksi}) \cdot \frac{3}{64} \text{ in} = 0.8 \text{ kip/in} \quad (3.28)$$

$$q_{\max} = \tau_{\max} \cdot t = (25.6 \text{ ksi}) \cdot \frac{3}{64} \text{ in} = 1.2 \text{ kip/in} \quad (3.29)$$

Using $q_{\max}$, we calculate the maximum load transfer for both orientations:

$$\text{Horizontal: Max load transfer} = 1.2 \text{ kip/in} \cdot \sec(42.5^\circ) = 1.6276 \text{ kip/in} \quad (3.30)$$

$$\text{Total max load transfer} = 1.6276 \text{ kip/in} \cdot 18 \text{ in} = 29.297 \text{ kip} \quad (3.31)$$

$$\text{Vertical: Max load transfer} = 1.2 \text{ kip/in} \cdot \csc(42.5^\circ) = 1.7762 \text{ kip/in} \quad (3.32)$$

$$\text{Total max load transfer} = 1.7762 \text{ kip/in} \cdot 10 \text{ in} = 17.762 \text{ kip} \quad (3.33)$$

From Table 2.3, the fastener material chosen is Ti-6Al-4V & Alloy Steel with a diameter of 5/32 in, and an ultimate single shear load of 1822 lb/fastener. The number of fasteners required is:

$$\text{Horizontal: } \frac{29.297 \text{ kip}}{1.822 \text{ kip/fastener}} = 16.08 \Rightarrow 17 \text{ fasteners} \quad (3.34)$$

$$\text{Vertical: } \frac{17.762 \text{ kip}}{1.822 \text{ kip/fastener}} = 9.75 \Rightarrow 10 \text{ fasteners} \quad (3.35)$$

For the actual load transfer, we use $q = 0.8 \text{ kip/in}$:

$$\text{Horizontal: Total load transfer} = 0.8 \text{ kip/in} \cdot \sec(42.5^\circ) \cdot 18 \text{ in} = 19.532 \text{ kip} \quad (3.36)$$

$$\text{Vertical: Total load transfer} = 0.8 \text{ kip/in} \cdot \csc(42.5^\circ) \cdot 10 \text{ in} = 11.842 \text{ kip} \quad (3.37)$$

The factors of safety for the rivets are:

$$FS_{V,\text{horizontal}} = \frac{17 \cdot 1.822 \text{ kip}}{19.532 \text{ kip}} = 1.59 \quad (3.38)$$

$$FS_{V,\text{vertical}} = \frac{10 \cdot 1.822 \text{ kip}}{11.842 \text{ kip}} = 1.54 \quad (3.39)$$

CONCLUSION

In this report, the design of the built-up main spar for an aircraft wing using 2024-T3 aluminum has been successfully completed. The skin thickness was determined based on shear stress calculations, and the stiffeners and flanges were selected by evaluating their load-bearing capacity and buckling resistance. The chosen components provide adequate strength with factors of safety above the required limit of 1.5, ensuring structural integrity under design loads. Additionally, the rivet arrangement was calculated to safely handle load transfers, meeting all necessary criteria for the given factor of safety. Overall, the design meets the requirements for both strength and stability.

| A

APPENDIX A

A.1 Rough Work

AerE 426 - Lab 4

① Skin design

$$V_{FS} = 1.5 \cdot 8000 \text{ lb} = 12000 \text{ lb}$$

$$\tau = \frac{V}{h \cdot t} \Rightarrow t_{min} = \frac{V_{FS}}{h \cdot \tau_{max}} = \frac{12000 \text{ lb}}{(10 \text{ in}) (37 \times 10^3 \text{ psi})} = 0.0324 \text{ in}$$

All sheets are in multiples of $\frac{1}{64} \text{ in} = 0.015625 \text{ in}$

$$\hookrightarrow t = \frac{3}{64} \text{ in} = 0.046875 \text{ in} \gg t_{min}$$

$$\tau = \frac{V}{h \cdot t} = \frac{8 \text{ kip}}{(10 \text{ in}) (\frac{3}{64} \text{ in})} = 17.067 \text{ ksi}$$

$$\begin{aligned} \tau_{max} &= \frac{V_{max}}{h \cdot t} \\ \tau_{max} &= \frac{12 \text{ kip}}{10 \text{ in} (\frac{3}{64} \text{ in})} \\ \tau_{max} &= 25.6 \text{ ksi} \end{aligned}$$

$$FS_{\tau} = \frac{\tau_{max}}{\tau} = \frac{37 \text{ ksi}}{17.0667 \text{ ksi}} = 2.17 \gg$$

② Design the flanges and stiffeners

* Stiffeners:

$$P_{cr} = \frac{V_{shear}}{h} \tan \alpha = \frac{12000 \text{ lb}}{(10 \text{ in})} \tan 42.5^\circ = 19.793 \text{ kip}$$

$$\sigma_{cr, max} = \frac{P_{cr}}{A_{min}} \Rightarrow A_{min} = \frac{P_{max}}{\sigma_{cr, max}} = \frac{19.793 \text{ kip}}{36 \text{ ksi}} = 0.550 \text{ in}^2 \quad (2 \times A)$$

$$I_{min} = \frac{P_{cr} L^2}{\pi^2 E} = \frac{(19.793 \text{ kip}) (10 \text{ in})^2}{\pi^2 (10.7 \times 10^3 \text{ ksi})} = 0.0187 \text{ in}^4$$

$$\begin{aligned} \text{Angle beam no 2: } A &= 0.376 \text{ in}^2 \times 2 = 0.752 \text{ in}^2 \\ I &= 0.0649 \text{ in}^4 \times 2 = 0.1298 \text{ in}^4 \end{aligned}$$

$$P = (18 \text{ kip}) \left(\frac{18 \text{ in}}{10 \text{ in}} \right) \tan(42.5^\circ) = 13.195 \text{ kip}$$

$$\sigma = \frac{P}{A_T} = \frac{13.195 \text{ kip}}{0.752 \text{ in}^2} = 17.547 \text{ ksi}$$

$$FS_{\sigma_c} = \frac{\sigma_{c, \text{yield}}}{\sigma} = \frac{36 \text{ ksi}}{17.547 \text{ ksi}} = 2.05_{,,}$$

$$P_{cr} = \frac{\pi^2 EI_T}{L^2} = \frac{\pi^2 (10.7 \times 10^3 \text{ ksi}) (0.0649 \text{ in}^4)}{(10 \text{ in})^2} = 137.075 \text{ kip}$$

$$FS_{\text{Buckling}} = \frac{P_{cr}}{P} = \frac{137.075 \text{ kip}}{13.195 \text{ kip}} = 10.4_{,,}$$

* Flange:

$$M_{fl} = \frac{\tau \tan \alpha \cdot t \cdot w^2}{12} = \frac{(17.067 \text{ ksi}) (\tan 42.5^\circ) (\frac{3}{64} \text{ in}) (18 \text{ in})^2}{12}$$

$$M_{fl} = 19.793 \text{ kip} \cdot \text{in}$$

$$P_{max} = \frac{V_{max} \cot \alpha}{2} = \frac{(12 \text{ kip}) \cot(42.5^\circ)}{2} = 6.548 \text{ kip}$$

$$M_{max} = \frac{\tau_{max} \tan \alpha \cdot t \cdot w^2}{12} = \frac{(25.6 \text{ ksi}) (\tan 42.5^\circ) (\frac{3}{64} \text{ in}) (18 \text{ in})^2}{12}$$

$$M_{max} = 29.689 \text{ kip} \cdot \text{in}$$

$$\sigma_{c, \text{yield}} = -36 \text{ ksi} = - \left(\frac{29.689 \text{ kip} \cdot \text{in}}{I} \right) + \frac{(12 \text{ kip}) \cot(42.5^\circ)}{2 \cdot A}$$

$$\sigma_{T, \text{yield}} = 44 \text{ ksi} = \left(\frac{29.689 \text{ kip} \cdot \text{in}}{I} \right) + \frac{(12 \text{ kip}) \cot(42.5^\circ)}{2 \cdot A}$$

$$\underline{I}_{\min} = 0.742 \text{ in}^3 \quad A = 1.636 \text{ in}^2 \quad \underline{I} = 1.9 \text{ in}^3$$

$$\text{Beam no 19: } A = 1.9 \text{ in}^2 \quad \underline{I} = 0.8339 \text{ in}^3 \quad \underline{I} = 3.4566 \text{ in}^4$$

$$c = \frac{\underline{I}}{I/c} = \frac{3.4566 \text{ in}^4}{0.8339 \text{ in}^3} = 4.1451 \text{ in}$$

$$\sigma_{fL,T} = \frac{M_{fL,C \& L}}{I_{fL}} + \frac{V \cot(\alpha)}{2A} = \frac{(19.793 \text{ kip} \cdot \text{in})}{(0.8339 \text{ in}^3)} + \frac{(8 \text{ kip}) \cot(42.5^\circ)}{2(1.9 \text{ in}^2)}$$

$$\sigma_{fL,T} = 26.033 \text{ ksi}$$

$$\sigma_{fL,C} = -\frac{M_{fL,C}}{I} + \frac{V \cot(\alpha)}{2A} = -21.438 \text{ ksi}$$

$$P_{fL} = \frac{V \cot \alpha}{2} = \frac{(8 \text{ kip}) \cot(42.5^\circ)}{2} = 4.3652 \text{ kip}$$

$$P_{fL,cr} = \frac{\pi^2 E I_{fL}}{L_{fL}^2} = \frac{\pi^2 (10.7 \times 10^3 \text{ ksi}) (3.4566 \text{ in}^4)}{(18 \text{ in})^2} = 1126.65 \text{ kip}$$

$$FS_{\sigma_T} = \frac{\sigma_{T,y}}{\sigma_{fL,T}} = \frac{44 \text{ ksi}}{26.033 \text{ ksi}} = 1.69$$

$$FS_{\sigma_C} = \frac{\sigma_{C,y}}{\sigma_{fL,C}} = \frac{36 \text{ ksi}}{21.438 \text{ ksi}} = 1.68$$

$$FS_{\text{buckling}} = \frac{P_{fL,cr}}{P_{fL}} = \frac{1126.65 \text{ kip}}{4.3652 \text{ kip}} = 258$$

(3) Rivet design

$$q_{\max} = \tau_{\max} \cdot t = (25.6 \text{ ksi}) \left(\frac{3}{64} \text{ in} \right) = 1.2 \text{ kip/in}$$

$$q = \tau \cdot t = (17.067 \text{ ksi}) \left(\frac{3}{64} \text{ in} \right) = 0.8 \text{ kip/in}$$

(flange) Horizontal: load transfer = $q \cdot \sec(\alpha) = (0.8 \text{ kip/in}) \sec(42.5^\circ) = 1.0851$
 . max load transfer = $q_{\max} \cdot \sec(\alpha) = (1.2 \text{ kip/in}) \sec(42.5^\circ) = 1.6276 \text{ kip/in}$
 total load transfer = $(1.0851 \text{ kip/in})(18 \text{ in}) = 19.532 \text{ kip}$
 total max load transfer = $(1.6276 \text{ kip/in})(18 \text{ in}) = 29.297 \text{ kip}$

(stiffener) Vertical: load transfer = $q \cdot \csc(\alpha) = (0.8 \text{ kip/in}) \csc(42.5^\circ) = 1.1842 \text{ kip/in}$
 . max load transfer = $q_{\max} \cdot \csc(\alpha) = (1.2 \text{ kip/in}) \csc(42.5^\circ) = 1.7762 \text{ kip/in}$
 total load transfer = $(1.1842 \text{ kip/in})(10 \text{ in}) = 11.842 \text{ kip}$
 total max load transfer = $(1.7762 \text{ kip/in})(10 \text{ in}) = 17.762 \text{ kip}$

Fasteners chosen: . Material: Ti-6Al-4V & Alloy steel
 . Diameter: 5/32 in
 . $V_{\text{ult}} = 1.822 \text{ kip/fastener}$
 . $T_u = 95 \text{ ksi}$

Horizontal: $\frac{29.297 \text{ kip}}{1.822 \text{ kip}} = 16.08 \Rightarrow 17 \text{ fasteners,}$

$FS_v = \frac{17 \cdot (1.822 \text{ kip})}{19.532 \text{ kip}} = 1.59$

(both) $FS_t = \frac{95 \text{ ksi}}{17.069 \text{ ksi}} = 5.57$

Vertical: $\frac{17.762 \text{ kip}}{1.822 \text{ kip}} = 9.75 \Rightarrow 10 \text{ fasteners}$

$FS_v = \frac{10 \cdot (1.822 \text{ kip})}{11.842 \text{ kip}} = 1.54$